

is not possible to use this physical board. Kanban board software tools are designed to manage such lightweight projects worked on by a distributed team. These tools are ideal for monitoring the progress of work through different stages, which can be viewed as a workflow.

Features of Kanban

Kanban originated in the automotive industry in Japan. Toyota developed a production system called just-in-time manufacturing (JIT) to improve manufacturing efficiency which used a Kanban board. Kanban in manufacturing is pull based, namely, driven by demand from downstream processes. Giving visibility to who owns each task, where each task is in the process stages, as well as keeping a tight check on work in progress (WIP) are all in-line with the Kanban used in manufacturing.

Visual and clear overview of tasks:

The Kanban board is the best way to know the current status of a project at a glance because it is visual. It's very easy to understand, there is nothing to explain and no training is required. Everyone can see who is doing what, the status of a task and whether any tasks are blocked.

Work in progress (WIP) can be capped: When teams juggle between multiple tasks, they lose focus and get distracted. Maybe your team is great at starting tasks but not so good at finishing them. You end up with many things partially done, very few completed and an increasingly confusing backlog of items that need to be addressed. Team members often waste some time and energy to pick up the thread where they left off and get back to the task after being distracted. The WIP limits prevent this wasted effort and energy by keeping new tasks out of the workflow until previous tasks have been fully completed.

Swimlanes for different work types: All work need not be brought into the same pool. You can have



Figure 1: Sticky notes showing tasks on a whiteboard

separate swimlanes for different types of work. For example, if you have some normal tasks and some priority tasks, you don't want the priority tasks to get buried under a lot of normal tasks. In such a situation, you can set up separate swimlanes for each type of task. Kanban swimlanes are horizontal bands that split a Kanban board into sections. You can also use them to visually separate, say, two sub-projects, within the larger project, on the same board and organise related tasks together.

Sub-tasks to show progress within a task: In the case of large tasks, it is not enough to know that the task is in progress. If you find out that a task is way behind schedule too close to the deadline, there may not be enough time to rectify that through emergency corrective actions. In proper project management tools that use a Gantt chart, this is handled by updating the task status with a percentage completion for each task. Kanban boards use a checklist of sub-tasks as a way of splitting a larger task into smaller chunks. The team member can keep crossing out completed items on the checklist, which gives a rough measure of how the task is progressing.

Adding comments and attaching documents to tasks: Team members may want to pass on information related to a task to the person doing that job. This will help the person who may have to test it later or help the supervisor too. The usual method of sending that information or attachment is by email, which is often difficult to trace. Instead of burying it in email, if the information is added as a comment or as an attachment to the task, it gives visibility to all concerned in a context-sensitive manner.

Connecting to your LDAP server for authentication: LDAP is the industry standard protocol for accessing and maintaining directory information of, say, all the employees of an organisation. Microsoft's Active Directory and OpenLDAP, for example, are LDAP-compliant directory services. When you log into your company's network, your user name and password are likely validated through this. If all team members are employees of your organisation, won't it be convenient if you can tap into your organisation's authentication system? The Kanban boards support connecting to your organisation's LDAP server for authentication.